

JOHN PAUL TAPUYO

AI Systems & Automation Engineer | Senior Flutter & Full-Stack Developer



Philippines



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PROFESSIONAL SUMMARY

AI Systems & Automation Engineer with 12+ years of software development experience and 7+ years specializing in Flutter and cross-platform applications. Experienced in designing and deploying LLM-powered applications, RAG pipelines, AI agents, and automation workflows using both cloud and local models. Proven ability to build production mobile/web apps, automate business processes, and integrate AI assistants into real-world systems. Strong background in startups, enterprise environments, and end-to-end product delivery.

CORE COMPETENCIES

AI ENGINEERING & LLM SYSTEMS

- LLM orchestration using LangChain & LangGraph
- Prompt Engineering & reasoning strategies:
 - Zero-shot, One-shot, Few-shot prompting
 - Chain-of-Thought reasoning workflows
 - Prompt evaluation and iterative optimization
- Retrieval-Augmented Generation (RAG)
 - Document ingestion, chunking & embedding pipelines
 - Context scoring, reranking, and relevance tuning
- Vector Databases & Semantic Search
 - Embeddings, similarity search, hybrid retrieval strategies
 - Metadata filtering and retrieval optimization
- AI Agent Workflows
 - Multi-step agents and tool usage
 - LangGraph orchestration vs traditional chains
 - Automation workflows connected to APIs and business systems
- Model Context Protocol (MCP) Tools
 - Connecting LLMs to external APIs, services and local tools
- LLM Deployment

- OpenAI, Gemini, Ollama, Gemma, local open-source models
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SOFTWARE ENGINEERING

- Flutter, FlutterFlow, React, Python, Golang, Node.js, Laravel, WordPress, PHP
 - REST & GraphQL API architecture
 - Android & iOS deployment pipelines
 - CI/CD: Codemagic, GitHub, GitLab, Vercel, Docker
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BACKEND & DATABASES

- Firebase, Supabase, MySQL, SQLite, AWS
 - Serverless and cloud integrations
 - Authentication, payments, real-time systems
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AUTOMATION & IoT

- n8n workflow automation, Make, Zapier
 - Arduino, ESP32, sensor integrations
 - AI-driven internal tools & business automation
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TOOLS

VSCode • Android Studio • Xcode • Antigravity • Claude Code • Cursor • Jupiter • Git • Copilot • Arduino IDE • Figma • Adobe XD • Photoshop • Canva

PROFESSIONAL EXPERIENCE

Ayowa Inc - Full Stack Developer

Jan 2026 – Present

Projects: Ayowa app

Tech: Flutter, NestJs, Docker, NextJs, Cockroach

- Create secured API endpoint
- Integration of SDK in backend such as payment
- Built mobile and web apps from scratch using clean architecture
- Implemented authentication, APIs, and email systems

- Managed Apple & Google developer accounts and releases

Skyshare Labs Inc — Senior Web & Mobile Developer / AI Engineer

Jun 2025 – Dec 2025

Projects: Zion IMS, Avian Summit, Awesome Smile

Tech: Flutter, Python, OpenAI, n8n, LLMs

- Designed and deployed RAG-powered AI assistants for business workflows and knowledge systems
 - Built multi-step AI agents connected to APIs and automation tools
 - Implemented prompt engineering and reasoning workflows for production use
 - Developed cross-platform mobile and web apps using Flutter
 - Architected databases, state management, and CI/CD pipelines
 - Launched apps on Web, Google Play, and Apple App Store
 - Built end-to-end automation workflows using n8n
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Innovative Global Entities — Full-Stack & AI Developer

Jan 2025 – May 2025

Projects: Innovative Website, BizCredit, Sales Flow

Tech: Flutter, Golang, Nest.js, React, NextJs, Shopify, LLMs, Twilio, LLM, Langchain

- Developed and maintained production mobile apps and backend systems
 - Built and deployed server-side AI assistants integrated into chat platforms
 - Integrated third-party APIs, payments, and automation workflows
 - Participated in sprint planning, branching strategies, and CI pipelines
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GECO PH / Unilab PH — Flutter Developer

Aug 2024 – Dec 2024

Projects: Pharma App, RiteMed, Alerta

Tech: Flutter, Unity, Python, AWS

- Delivered enterprise Flutter features based on sprint tickets
- Managed deployment and release cycles
- Collaborated with clients and stakeholders in production environments

Prodigy Works — Full-Stack Developer

Oct 2023 – Jun 2024

Projects: Examathleet, Knot Dating, St Lukes Medical HR App

Tech: Flutter, Go, React, Firebase, Supabase

- Built mobile and web apps from scratch using clean architecture
- Implemented authentication, APIs, and email systems
- Managed Apple & Google developer accounts and releases

Theoria Medical Institute (US) — Senior Flutter Developer

Dec 2021 – Oct 2023

Projects: CharEasy

Tech: Flutter, GraphQL, Firebase

- Led feature development and code quality reviews
- Mentored junior developers and enforced best practices
- Implemented CI/CD pipelines and TestFlight deployments

New Driver Programme (UK) — Senior Full-Stack Developer

Dec 2020 – Dec 2021

Tech: Driving Lesson Route, NDP Website

Tech: React, Flutter, Firebase, Google Cloud, Wordpress, Nodejs

- Architected mobile apps and Firebase backend
- Integrated mapping solutions (Mapbox, HERE Maps)
- Supported WordPress and BuddyBoss web systems

Leepe Inc — Senior Full-Stack Developer / Team Lead

Mar 2020 – Nov 2020

Tech: DrinkLink, Vibelynx

Tech: Flutter, ReactJs, NestJs, Supabase, Firebase, Django

- Led development team and conducted code reviews
 - Managed client meetings and technical planning
 - Supervised deployments and releases
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Eco Access Solution — IT Head

Nov 2019 – Mar 2020

- Developed internal systems and employee monitoring tools
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Bohol Island State University — IT Head / Programmer

Jul 2018 – Jun 2019

- Built IoT automation systems using sensors
 - Developed desktop apps for data visualization
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SELECTED AI & AUTOMATION PROJECTS

Local AI Business Assistant

- Built local LLM assistant using Ollama integrated with Telegram and workflows
- Implemented knowledge-based RAG system for business operations

AI Chat Automation Systems

- Developed AI chat tools connected to APIs and internal workflows

IoT + Mobile Integration Projects

- Oil Engine Quality Tester (ESP32 + Flutter Bluetooth app)
 - Smart irrigation system using Arduino and soil sensors
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EDUCATION

Bachelor of Science in Computer Science
Bohol Island State University — 2012–2018

AREAS OF EXPERTISE

AI Systems & LLM Applications • AI Agents • RAG Pipelines • Automation • Mobile Development
• Web Development • Backend Systems • IoT • UI/UX

REFERENCE

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